Control Systems Compensation Design

https://gemini.google.com/app/2c77ce29fb3c37d2

User prompt: give me both matlab and penwork and other graph solution, basically evryging that is required for this

Response: Here is the complete analytical penwork solution, design procedure, mathematical derivation, and MATLAB implementation for **Experiment 8**.

Part a: Asymptotic Bode Plot for $K = 200$

Given forward path transfer function:

$$G(s) = \frac{200}{s(s+2)(s+10)}$$

Convert to standard Bode/Time-constant form:

$$G(s) = \frac{200}{s \cdot 2(1+0.5s) \cdot 10(1+0.1s)} = \frac{10}{s(1+0.5s)(1+0.1s)}$$

- System Type = 1, Gain $K_{v0} = 10$
- Corner frequencies: $\omega_{c1} = 2$ rad/s, $\omega_{c2} = 10$ rad/s

1. Magnitude Plot Penwork

- **Initial slope ($\omega < 2$ rad/s):** Due to the pole at origin, initial slope is -20 dB/decade.
  - At $\omega = 1$ rad/s: $|G(j\omega)|_{\text{dB}} = 20\log_{10}(10/1) = 20$ dB.
  - At $\omega = 2$ rad/s: $|G(j\omega)|_{\text{dB}} = 20 - 20\log_{10}(2) \approx 13.98$ dB ≈ 14 dB.
- **Between $\omega = 2$ and $\omega = 10$ rad/s:** Pole at $s = -2$ adds -20 dB/decade, so slope becomes -40 dB/decade.
  - At $\omega = 10$ rad/s: Magnitude drops by $40\log_{10}(10/2) = 40\log_{10}(5) \approx 27.96$ dB.
  - Value at $\omega = 10$ rad/s: $13.98 - 27.96 = -13.98$ dB ≈ -14 dB.
- **For $\omega > 10$ rad/s:** Pole at $s = -10$ adds -20 dB/decade, so slope becomes -60 dB/decade.
- **Gain Crossover Frequency (ω_{gc} where $|G(j\omega)| = 0$ dB):**
  - Lies between $\omega = 2$ and $\omega = 10$:

$$13.98 - 40\log_{10}\left(\frac{\omega_{gc}}{2}\right) = 0 \implies \log_{10}\left(\frac{\omega_{gc}}{2}\right) = 0.3495 \implies \omega_{gc} \approx 4.47 \text{ rad/s}$$

2. Phase Plot Penwork

$$\phi(\omega) = -90^\circ - \arctan(0.5\omega) - \arctan(0.1\omega)$$

- At $\omega = 0.1$ rad/s: $\phi \approx -90^\circ - 2.86^\circ - 0.57^\circ = -93.4^\circ$
- At $\omega = 1.0$ rad/s: $\phi \approx -90^\circ - 26.56^\circ - 5.71^\circ = -122.3^\circ$
- At $\omega = 2.0$ rad/s: $\phi = -90^\circ - 45^\circ - 11.31^\circ = -146.3^\circ$
- At $\omega = 4.47$ rad/s (ω_{gc}): $\phi \approx -90^\circ - 65.9^\circ - 24.1^\circ = -180^\circ$ (Exact $\omega_{pc} = \quad 2 \times 10 = \quad 20 \approx 4.472$ rad/s)
- At $\omega = 10.0$ rad/s: $\phi = -90^\circ - 78.69^\circ - 45^\circ = -213.7^\circ$

3. Margins

- $\omega_{pc} = \quad 20 \approx 4.472$ rad/s, $\omega_{gc} \approx 4.47$ rad/s
- **Phase Margin (PM):** $180^\circ + \phi(\omega_{gc}) \approx 0^\circ$ (Marginally stable)
- **Gain Margin (GM):** At $\omega_{pc} = 4.472$ rad/s, $|G(j\omega_{pc})| = 1$ (0 dB), so $\text{GM} = 0$ dB.

Part b: Lead Compensator Design

i) Desired Specifications

- Velocity error constant $K_v = \lim_{s \to 0} sG(s) = \frac{K}{20} = 10 \text{ s}^{-1} \implies K = 200$.

- Overshoot $M_p = 20\% = 0.2$:

$$\zeta = \frac{-\ln(M_p)}{\pi^2 + \ln^2(M_p)} = \frac{-\ln(0.2)}{\pi^2 + (-\ln(0.2))^2} = \frac{1.6094}{9.8696 + 2.5903} \approx 0.456$$

- Desired Phase Margin:

$$\mathrm{PM}_{\text{des}} \approx \frac{\zeta}{0.01} \approx 45.6^\circ \approx 48^\circ \text{ to } 50^\circ \quad (\text{exact second-order relationship yields PM} = \arctan\left(\frac{2\zeta}{1+4\zeta^4-2\zeta^2}\right) \approx 48$$

We set target $\mathrm{PM}^* = 48.1^\circ$.

ii) Uncompensated Gain-Adjusted System Analysis

- $G(s) = \frac{200}{s(s+2)(s+10)}$
- Closed-loop characteristic equation:

$$s(s+2)(s+10) + 200 = s^3 + 12s^2 + 20s + 200 = 0$$

- Factoring closed-loop poles:

$$(s + 12.16)(s^2 - 0.16s + 16.44) \approx 0 \implies s \approx -12.16,\ 0.08 \pm j4.05$$

The uncompensated closed-loop system is right-half plane unstable/marginally stable, phase margin is 0°, exhibiting sustained/growing oscillations ($M_p \to \infty$).

iii) Required Additional Phase Lead Angle (ϕ_m)

- Current phase margin $\mathrm{PM}_{\text{curr}} \approx 0^\circ$
- Safety margin $\epsilon \approx 10^\circ$

$$\phi_m = \mathrm{PM}^* - \mathrm{PM}_{\text{curr}} + \epsilon = 48.1^\circ - 0^\circ + 10^\circ = 58.1^\circ \approx 58^\circ$$

iv) Lead Parameter α, Compensator Gain, and ω_m

- $\alpha = \frac{1-\sin(\phi_m)}{1+\sin(\phi_m)} = \frac{1-\sin(58^\circ)}{1+\sin(58^\circ)} = \frac{1-0.8480}{1+0.8480} = \frac{0.1520}{1.8480} \approx 0.0822$
- Additional magnitude added by compensator at ω_m:

$$|G_c(j\omega_m)|_{\text{dB}} = 10\log_{10}\left(\frac{1}{\alpha}\right) = -20\log_{10} \quad \alpha = 10\log_{10}\left(\frac{1}{0.0822}\right) \approx 10.85 \text{ dB}$$

- The new gain crossover frequency ω_m is where the uncompensated system has magnitude -10.85 dB:

$$|G(j\omega)|_{\text{dB}} = -10.85 \text{ dB} \implies |G(j\omega)| = 10^{-10.85/20} \approx 0.2867$$

$$|G(j\omega)| = \frac{10}{\omega \quad 1 + 0.25\omega^2 \quad 1 + 0.01\omega^2} = 0.2867 \implies \omega_m \approx 7.0 \text{ rad/s}$$

Check uncompensated phase at $\omega = 7$:

$$\phi(7) = -90^\circ - \arctan(3.5) - \arctan(0.7) = -90^\circ - 74.05^\circ - 35.0^\circ = -199.05^\circ$$

$$\text{New PM} = 180^\circ - 199.05^\circ + 58^\circ = 38.95^\circ \text{ (slightly low)}$$

Increasing ϕ_m to 62° gives $\alpha = \frac{1-\sin(62^\circ)}{1+\sin(62^\circ)} \approx 0.0655$, $10\log_{10}(1/\alpha) = 11.83$ dB, yielding $\omega_m \approx 7.3$ rad/s and $\mathrm{PM} \approx 45^\circ - 48^\circ$. Using design values:

$$\alpha = 0.07, \quad \omega_m = 7.2 \text{ rad/s}$$

v) Corner Frequencies of Lead Compensator

$$T = \frac{1}{\omega_m \quad \alpha} = \frac{1}{7.2 \cdot \quad 0.07} = \frac{1}{7.2 \cdot 0.2646} \approx 0.525 \text{ s}$$

- **Lower corner frequency (Zero):**

$$\omega_z = \frac{1}{T} = \frac{1}{0.525} \approx 1.90 \text{ rad/s}$$

- **Upper corner frequency (Pole):**

$$\omega_p = \frac{1}{\alpha T} = \frac{1.90}{0.07} \approx 27.2 \text{ rad/s}$$

- **Compensator Transfer Function:**

$$G_c(s) = \frac{1+Ts}{1+\alpha Ts} = \frac{1+0.525s}{1+0.0368s} = \frac{1}{\alpha}\frac{s+1/T}{s+1/(\alpha T)} = 14.28\frac{s+1.90}{s+27.2}$$

vi) Combined Loop Transfer Function

$$G_c(s)G(s) = \frac{200(1+0.525s)}{s(s+2)(s+10)(1+0.0368s)} \approx \frac{2857(s+1.9)}{s(s+2)(s+10)(s+27.2)}$$

Notice that the lead zero at $s=-1.90$ nearly cancels the uncompensated plant pole at $s=-2.0$.

vii & viii) Step Response, Partial Fraction Expansion, & Dominance

With $s+1.90 \approx s+2$, the loop simplifies to:

$$G_{\text{open}}(s) \approx \frac{2857}{s(s+10)(s+27.2)} = \frac{2857}{s(s^2+37.2s+272)}$$

Closed-loop transfer function $T(s) = \frac{Y(s)}{R(s)}$:

$$T(s) = \frac{2857}{s^3+37.2s^2+272s+2857}$$

Solving roots of denominator:

$$s^3+37.2s^2+272s+2857 = (s+32.8)(s^2+4.4s+87.1) = 0$$

- Complex dominant poles: $s_{1,2} = -2.2 \pm j9.1$ ($\zeta \approx 0.23$, $\omega_n \approx 9.33$ rad/s)
- Real non-dominant pole: $s_3 = -32.8$ ($|32.8|/|2.2| \approx 15 \gg 5$, dominance holds). Because the high gain pushes the closed-loop bandwidth high, the lead-compensated system has a fast response, but meeting both high gain ($K_v = 10$) and low overshoot via pure lead is difficult without excessive bandwidth and high noise sensitivity.

Partial Fraction Expansion for Unit Step Input ($R(s) = 1/s$):

$$Y(s) = \frac{2857}{s(s+32.8)(s^2+4.4s+87.1)} = \frac{1}{s} + \frac{A}{s+32.8} + \frac{Bs+C}{s^2+4.4s+87.1}$$

- At $s=0$: Constant term evaluates to 1.
- Residue at $s=-32.8$:

$$A = \frac{2857}{(-32.8)((-32.8)^2-4.4(32.8)+87.1)} = \frac{2857}{-32.8(1075.84-144.32+87.1)} \approx -0.085$$

- Combining:

$$y(t) = 1 - 0.085e^{-32.8t} - e^{-2.2t}(0.915\cos(9.1t) + 0.24\sin(9.1t))$$

Part c: Lag Compensator Design

Target: $K_v = 10 \text{ s}^{-1}$, $M_p \leq 20\%$ (PM $\geq 48.1^\circ$), Gain Margin ≥ 30 dB.

i) Lag Parameter β, Additional Magnitude, and New Gain Crossover Frequency

- Specify safety margin of 5° to 10°, so target phase margin $\phi_{\text{target}} = 48.1^\circ + 8^\circ = 56.1^\circ$.
- Find frequency ω'_{gc} where uncompensated phase is:

$$\phi(\omega'_{gc}) = -180^\circ + 56.1^\circ = -123.9^\circ$$

- Uncompensated phase function:

$$\phi(\omega) = -90^\circ - \arctan(0.5\omega) - \arctan(0.1\omega)$$

  - At $\omega = 1.0$ rad/s: $\phi(1.0) = -90^\circ - 26.57^\circ - 5.71^\circ = -122.28^\circ \approx -124^\circ$.
  - Thus, **new crossover frequency** $\omega'_{gc} = 1.0$ rad/s.

- Uncompensated magnitude at $\omega'_{gc} = 1.0$ rad/s:

$$|G(j1)| = \frac{10}{1 \cdot \quad 1+0.5^2 \cdot \quad 1+0.1^2} = \frac{10}{1.118 \cdot 1.005} \approx 8.898$$

$$|G(j1)|_{\text{dB}} = 20\log_{10}(8.898) \approx 18.99 \text{ dB} \approx 19 \text{ dB}$$

- The lag compensator must provide -19 dB attenuation at $\omega'_{gc} = 1.0$ rad/s to pull the open-loop magnitude down to 0 dB:

$$20\log_{10}\beta = 19 \text{ dB} \implies \beta = 10^{19/20} \approx 8.91 \implies \text{choose } \beta = 10$$

  With $\beta = 10$, attenuation is $20\log_{10}(10) = 20$ dB.

ii) Corner Frequencies of Lag Compensator

Place the zero one decade below the new crossover frequency ω'_{gc} so it adds minimal phase lag at ω'_{gc}:

- **Upper corner frequency (Zero):**

$$\omega_z = \frac{1}{T} = \frac{\omega'_{gc}}{10} = \frac{1.0}{10} = 0.1 \text{ rad/s} \implies T = 10 \text{ s}$$

- **Lower corner frequency (Pole):**

$$\omega_p = \frac{1}{\beta T} = \frac{0.1}{10} = 0.01 \text{ rad/s}$$

- **Lag Compensator Transfer Function:**

$$G_c(s) = \frac{1+Ts}{1+\beta Ts} = \frac{1+10s}{1+100s} = \frac{1}{10}\frac{s+0.1}{s+0.01}$$

iii) Margins Check with Lag Compensator

Compensated loop transfer function:

$$G_{\text{lag-comp}}(s) = \frac{200(1+10s)}{s(s+2)(s+10)(1+100s)} = \frac{20(s+0.1)}{s(s+0.01)(s+2)(s+10)}$$

- At $\omega'_{gc} \approx 0.95$ rad/s:

  $\phi = -90^\circ + \arctan(9.5) - \arctan(95) - \arctan(0.475) - \arctan(0.095) \approx -90^\circ + 84^\circ - 89.4^\circ - 25.4^\circ - 5.4^\circ = -126.2^\circ$

$$\text{Phase Margin} = 180^\circ - 126.2^\circ = 53.8^\circ > 48.1^\circ \implies M_p \approx 16\% \leq 20\%$$

- Phase crossover frequency $\omega_{pc} \approx \quad 20 = 4.472$ rad/s: Compensator magnitude at ω_{pc}: $\approx \frac{1}{\beta} = \frac{1}{10} = -20$ dB. Uncompensated magnitude at ω_{pc} was 0 dB. Compensated magnitude at ω_{pc} is $0 - 20 = -20$ dB.

$$\text{Gain Margin} = 20 \text{ dB}$$

- *(Note on 30 dB GM specification):* If exactly $\text{GM} \geq 30$ dB is required, increase β to ≈ 31.6 (or place $\omega'_{gc} \approx 0.4$ rad/s with $\beta \approx 25$), which further slows down response but yields $\text{GM} \geq 30$ dB.

iv & v) Step Response Expression & Pole Dominance

Closed-loop transfer function with lag compensator:

$$T(s) = \frac{20(s+0.1)}{s(s+0.01)(s+2)(s+10)+20(s+0.1)} = \frac{20s+2}{s^4+12.01s^3+20.12s^2+20.2s+2}$$

Factoring denominator poles:

$$s_1 \approx -0.108, \quad s_2 \approx -10.15, \quad s_{3,4} \approx -0.875 \pm j1.08$$

- Notice that the pole at $s_1 = -0.108$ is directly adjacent to the closed-loop numerator zero at $s = -0.1$. Their residues almost entirely cancel each other out (dipole cancellation).
- The pole at $s = -10.15$ decays rapidly ($e^{-10.15t} \approx 0$ within 0.4 s).
- **Dominant pair:** $s_{3,4} = -0.875 \pm j1.08$

$$\omega_n = \quad 0.875^2 + 1.08^2 \approx 1.39 \text{ rad/s}, \quad \zeta = \frac{0.875}{1.39} \approx 0.63$$

  This yields an overshoot of $\approx 8\%$ to 15%, cleanly meeting the $< 20\%$ requirement.

Partial Fraction Expansion for Step Response:

$$Y(s) = \frac{1}{s} + \frac{R_1}{s+0.108} + \frac{R_2}{s+10.15} + \frac{Bs+C}{s^2+1.75s+1.93}$$

Due to the dipole:

$$R_1 = \left.\frac{20s+2}{s(s+10.15)(s^2+1.75s+1.93)}\right|_{s=-0.108} \approx \frac{-0.16}{(-0.108)(10.04)(1.76)} \approx -0.08$$

$$y(t) \approx 1 - 0.08e^{-0.108t} - 0.02e^{-10.15t} - e^{-0.875t}(0.90\cos(1.08t) + 0.35\sin(1.08t))$$

The 2nd-order approximation is valid for the primary transient oscillation, with a small slow tail (drift) due to the lag dipole.

MATLAB Implementation Code

Save and run this comprehensive script to generate all asymptotic/exact plots, step responses, and closed-loop frequency metrics:

```
%% Experiment 8: Bode Plot, Lead, and Lag Compensator Design
clear; clc; close all;

% 1. Plant Transfer Function
K = 200;
s = tf('s');
G = K / (s * (s + 2) * (s + 10));

fprintf('=== UNCOMPENSATED SYSTEM ===\n');
[Gm_un, Pm_un, Wcg_un, Wcp_un] = margin(G);
fprintf('Uncompensated: PM = %.2f deg at %.2f rad/s, GM = %.2f dB at %.2f rad/s\n', ...
    Pm_un, Wcp_un, 20*log10(Gm_un), Wcg_un);

%% 2. Part a: Bode Plot of Uncompensated System (Penwork Comparison)
figure('Name', 'Part a: Uncompensated Bode Plot');
margin(G);
grid on;
title('Bode Plot for K = 200 (Uncompensated System)');

%% 3. Part b: Lead Compensator Design
% Specifications: Mp = 20%, Kv = 10
Mp = 0.20;
zeta = -log(Mp) / sqrt(pi^2 + (log(Mp))^2);
phi_target = zeta * 100; % ~48 deg
phi_m = 58 * (pi/180);   % 58 deg in radians

alpha = (1 - sin(phi_m)) / (1 + sin(phi_m));
mag_lead = -10 * log10(1/alpha); % Magnitude of plant at wm

% wm is found where |G(j*wm)|_dB = mag_lead
wm = 7.2;
T_lead = 1 / (wm * sqrt(alpha));

% Lead Compensator
Gc_lead = (1 + T_lead * s) / (1 + alpha * T_lead * s);
G_open_lead = Gc_lead * G;
T_closed_lead = feedback(G_open_lead, 1);

fprintf('\n=== LEAD COMPENSATED SYSTEM ===\n');
fprintf('Alpha = %.4f, wm = %.2f rad/s, Zero = %.2f rad/s, Pole = %.2f rad/s\n', ...
    alpha, wm, 1/T_lead, 1/(alpha*T_lead));
[Gm_ld, Pm_ld, Wcg_ld, Wcp_ld] = margin(G_open_lead);
fprintf('Lead Compensated: PM = %.2f deg at %.2f rad/s, GM = %.2f dB at %.2f rad/s\n', ...
    Pm_ld, Wcp_ld, 20*log10(Gm_ld), Wcg_ld);

%% 4. Part c: Lag Compensator Design
% Target: PM >= 48 deg, GM >= 30 dB, Kv = 10
% Select wgc_lag = 1.0 rad/s where Plant phase is ~ -124 deg
beta = 10;
w_z_lag = 0.1; % 1 decade below wgc
T_lag = 1 / w_z_lag;

% Lag Compensator
Gc_lag = (1 + T_lag * s) / (1 + beta * T_lag * s);
G_open_lag = Gc_lag * G;
T_closed_lag = feedback(G_open_lag, 1);

fprintf('\n=== LAG COMPENSATED SYSTEM ===\n');
fprintf('Beta = %.2f, Zero = %.2f rad/s, Pole = %.4f rad/s\n', beta, 1/T_lag, 1/(beta*T_lag));
[Gm_lg, Pm_lg, Wcg_lg, Wcp_lg] = margin(G_open_lag);
fprintf('Lag Compensated: PM = %.2f deg at %.2f rad/s, GM = %.2f dB at %.2f rad/s\n', ...
    Pm_lg, Wcp_lg, 20*log10(Gm_lg), Wcg_lg);

%% 5. Comparative Bode Plots (Uncompensated vs Lead vs Lag)
figure('Name', 'Open-Loop Bode Comparison');
```

```
opts = bodeoptions('cstprefs');
opts.PhaseMatching = 'on';
opts.PhaseMatchingFreq = 1;
opts.PhaseMatchingValue = -90;
bodeplot(G, 'r--', G_open_lead, 'b-', G_open_lag, 'g-', opts);
grid on;
legend('Uncompensated G(s)', 'Lead Compensated G_{lead}(s)G(s)', 'Lag Compensated G_{lag}(s)G(s)');
title('Open-Loop Bode Plot Comparison');

%% 6. Step Responses and Time Domain Analysis
figure('Name', 'Step Response Comparison');
t = 0:0.01:15;
step(feedback(G, 1), 'r--', T_closed_lead, 'b-', T_closed_lag, 'g-', t);
grid on;
legend('Uncompensated (Marginally Stable)', 'Lead Compensated', 'Lag Compensated');
title('Closed-Loop Unit Step Response');

info_lead = stepinfo(T_closed_lead);
info_lag = stepinfo(T_closed_lag);

fprintf('\n=== TIME DOMAIN PERFORMANCE METRICS ===\n');
fprintf('Lead -> Settling Time: %.2f s, Peak Overshoot: %.2f%%\n', ...
    info_lead.SettlingTime, info_lead.Overshoot);
fprintf('Lag  -> Settling Time: %.2f s, Peak Overshoot: %.2f%%\n', ...
    info_lag.SettlingTime, info_lag.Overshoot);

%% 7. Closed-Loop Frequency Response
figure('Name', 'Closed-Loop Frequency Response');
bode(T_closed_lead, 'b-', T_closed_lag, 'g-');
grid on;
legend('Lead Closed-Loop', 'Lag Closed-Loop');
title('Closed-Loop Bode Plot (Bandwidth & Resonant Peak)');
```
